TinyAudit: On-Device, Uncertainty-Aware Fairness Audits that Survive (and Expose) Compression

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Abstract

Machine learning is moving onto microcontroller-class hardware, where models are aggressively compressed (pruned, quantized) to fit. Responsible-AI audits, in contrast, are almost always run on full-precision models on a workstation, and they typically look at fairness, uncertainty, and explainability in isolation. We present TinyAudit, an auditing pipeline that measures all three together on the compressed artifact rather than the original, and that profiles its own cost. Theory already tells us calibration and parity cannot generally agree, and that compression harms underrepresented groups; we claim neither as new. Our contribution is measuring what those results do to a deployed model, reported as means over 10 seeds. First, the two lenses do not merely disagree, they select different groups: the ordering by selection rate and the ordering by per-group calibration error cross on UCI Adult and again on COMPAS, whose favorable direction is inverted. Second, and more actionable, compression moves the two in opposite directions: pruning an MLP to 90% sparsity drives its demographic-parity difference down roughly 12-fold while per-group calibration error climbs roughly 10-fold, so a parity-only audit run after compression ranks the most degraded model in the sweep as the fairest. Third, the audit's own peak working set is about 0.6 MB plus 2.3 KB per sample, constant in test-set size when streamed. That constant does not yet fit the 32 to 512 KB parts at the low end of the TinyML range, and we report the gap rather than claim it closed.

1 Introduction

TinyML runs inference on microcontrollers with 32 to 512 KB of SRAM [19]. To fit that budget, deployed models are pruned and quantized. The auditing tools that would check those models for fairness were built for a different world: cloud notebooks, full-precision weights, and demographic labels available at test time [14, 2, 8]. Existing TinyML benchmarks measure latency, accuracy, and energy, but not fairness or explainability [19]. The result is a gap. The models most likely to be compressed are the least likely to be audited, and when they are audited, the audit sees a model that is not the one deployed.

A second problem is that fairness audits usually stop at point predictions. A model can look fair on selection rates while its uncertainty is biased against a group [13], and abstaining on high-uncertainty inputs does not by itself repair group imbalance [17]. Uncertainty quantification now fits on KB-sized devices [7], but no existing system evaluates those estimates through a fairness lens, and no fairness toolkit fits on the device.

TinyAudit closes both gaps at once. It is a seven-stage pipeline that measures fairness, uncertainty, and explainability on the compressed model, inside a memory envelope sized for the device being audited. Our thesis: compression-era audits must be uncertainty-aware, because point fairness and calibration fairness are decoupled, compression can move one without the other, and all of this can be measured inside the device's own compute budget.

Relation to known results. We want to be precise about what is and is not new here, because both of our empirical results have theoretical ancestors. That calibration and parity-style criteria cannot generally be satisfied at once is established: Kleinberg et al. [11] proved the incompatibility for risk scores, Chouldechova [4] proved it for recidivism prediction on COMPAS specifically, and Pleiss et al. [16] characterized what calibration costs on the other criteria. That compression disproportionately damages underrepresented groups is likewise documented, most directly by Hooker et al. [9] through compression-identified exemplars, and by Blakeney et al. [3] for pruning specifically. We claim neither result as new, and a reader who knows these papers should expect our qualitative direction in advance.

What is new is the measurement setting. Those results were established on full-precision models with workstation tooling, and the model actually deployed to a microcontroller is neither of those things. Our claim is that the two literatures interact in a way that matters operationally: compression moves point parity and calibration fairness in *opposite* directions, so the metric most audits report improves as the model gets worse, and no existing toolkit measures this on the compressed artifact inside the device's own budget. The contribution is an instrument that makes that interaction visible, plus the evidence that it needs to be.

Contributions:

1. An on-device audit instrument: a seven-stage pipeline coupling fairness, uncertainty, and explainability, measured on the compressed model, emitting a one-page traffic-light card and a machine-readable manifest.
2. A measurement showing the calibration/parity tension is group-selective in practice on the deployed model: the two lenses rank sensitive groups differently on Adult and replicate on COMPAS, whose favorable direction is inverted (Section 4).
3. A concrete failure mode for compression-era audits: under pruning, demographic parity falls roughly 12-fold while per-group calibration error rises roughly 10-fold, so a parity-only audit ranks the most degraded model as the fairest (Section 5).
4. A feasibility measurement of the audit itself, which to our knowledge has not been reported before: peak working set is about 0.6 MB plus 2.3 KB per sample and is constant in test-set size when streamed (Section 6).

2 The TinyAudit instrument

TinyAudit takes a fitted model and a labeled test set and produces a one-page audit card plus a JSON manifest that every stage writes to. The card is generated from the manifest and never hand-edited; every figure in it is backed by a committed CSV. The stages, in order:

1. Profile. Parameter count, an analytic FLOPs proxy, peak RAM (via `tracemalloc`), and wall-clock per sample. **2. Point fairness.** Demographic-parity difference (DP), equalized-odds difference (EO), and disparate-impact ratio (DI), each with an explicitly pinned favorable direction and a hypothesis-tested range. **3. Uncertainty.** MC dropout (MLP only), a lightweight perturbation ensemble, and a QUTE-style early-exit estimator [7]; each returns predictive entropy, variance, and mutual information. **4. Uncertainty-aware fairness.** Per-group predictive entropy, per-group expected calibration error (ECE), and a selective-fairness AUC that integrates parity as low-confidence predictions are progressively abstained. **5. Explainability.** SHAP, LIME, and a dependency-free occlusion explainer, cross-checked for top-feature flips across sensitive groups. SHAP and LIME are costly and fragile at the edge [18], and embedded XAI needs design choices general-purpose toolkits do not automate [15], which is why the occlusion explainer ships alongside them. **6. Compression.** int8 dynamic quantization or magnitude pruning, applied *before* every metric above, so the audit sees the deployed model. **7. Render.** The one-page, traffic-light-coded card.

One design constraint shapes everything: the memory envelope. The audit's working set is dominated by per-sample buffers, so it can be streamed in small batches at a peak-RAM cost independent of test-set size (Section 6). The whole public API is one call:

```
from tinyaudit import audit
```

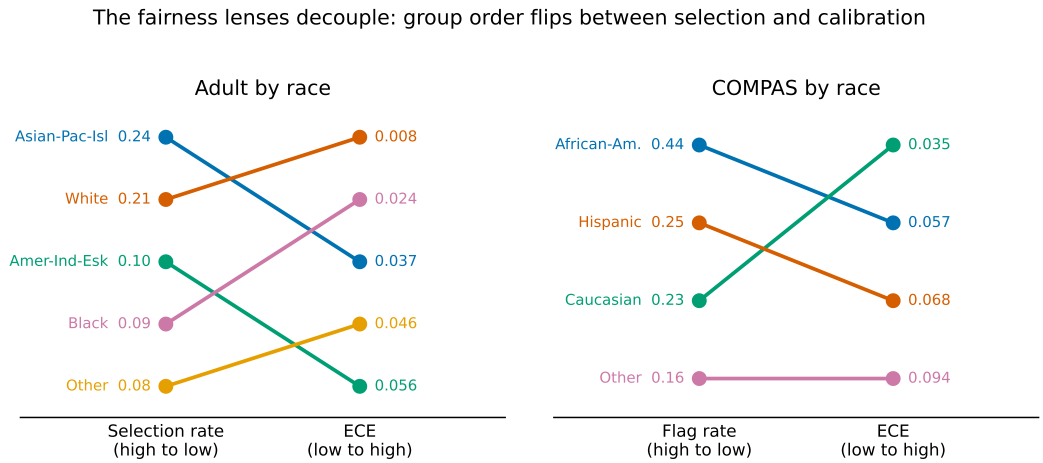

Figure 1: Finding 1. Rank slope charts for Adult (left) and COMPAS (right) by race, uncompressed logistic model, means over 10 seeds. Each line joins a group's rank by selection/flag rate to its rank by per-group ECE. Crossing lines mean the two lenses order the groups differently.

```
card = audit(model=clf, data=(X_test, y_test), sensitive="sex",
             methods=["fairness", "uncertainty", "xai"], compression="int8")
card.to_pdf("adult_lr_int8.pdf")
```

3 Experimental setup

Datasets. UCI Adult [12] (36,631 train / 12,211 test, 101 features after preprocessing; sensitive attributes sex and race), COMPAS [1] (4,629 / 1,543, 5 features; on COMPAS the positive outcome is being flagged high-risk, so higher is worse), and the ACSIncome subset of Folktables [5] (146,748 / 48,917, 8 features, 2018 California). Adult and COMPAS are the near-universal fairness benchmarks [6]; ACSIncome is the modern replacement for Adult's 1994 distribution [5].

Models. Standard scikit-learn estimators fit on standardized features: logistic regression, an MLP with two hidden layers of 32 units, and a decision tree. Scaling matters: without it the Adult MLP sits near the majority-class rate (about 0.54 accuracy) versus about 0.84 with it. The scaler is fit on train only and applied outside the model, so compression and perturbation reach the raw weights.

Reproducibility. A single seed flag seeds `random`, NumPy, torch, and `PYTHONHASHSEED`. Every headline number is a mean $\pm$ std over seeds 0 to 9. Determinism is guaranteed on Linux/x86; elsewhere we expect last-digit float drift, which the std columns absorb. All models land in published ranges: Adult logistic regression reaches accuracy 0.852 ± 0.003 (ROC-AUC 0.906 ± 0.002) and the Adult MLP 0.837 ± 0.003 (0.884 ± 0.003); on COMPAS the logistic model reaches 0.680 ± 0.006 (0.731 ± 0.008) and the MLP 0.683 ± 0.009; on ACSIncome the MLP reaches 0.808 ± 0.003 (0.889 ± 0.002). The fairness findings therefore sit on top of models that work, not on degenerate predictors.

4 Result 1: the two lenses select different groups

Point-prediction fairness asks how often each group receives the favorable outcome. Calibration fairness asks how honest the model's confidence is per group. The impossibility results [11, 4] say these cannot generally agree; what they do not say is *which* group each lens singles out on a given deployed model, and that is what an auditor actually has to act on. On the uncompressed logistic model, the two lenses rank the sensitive groups differently, and the disagreement is not a uniform offset but a reordering (Figure 1).

Adult by race (five groups). By selection rate the ordering is Asian-Pacific-Islander (0.240 ± 0.014), White (0.207 ± 0.005), American-Indian-Eskimo (0.098 ± 0.026), Black (0.091 ± 0.008), Other

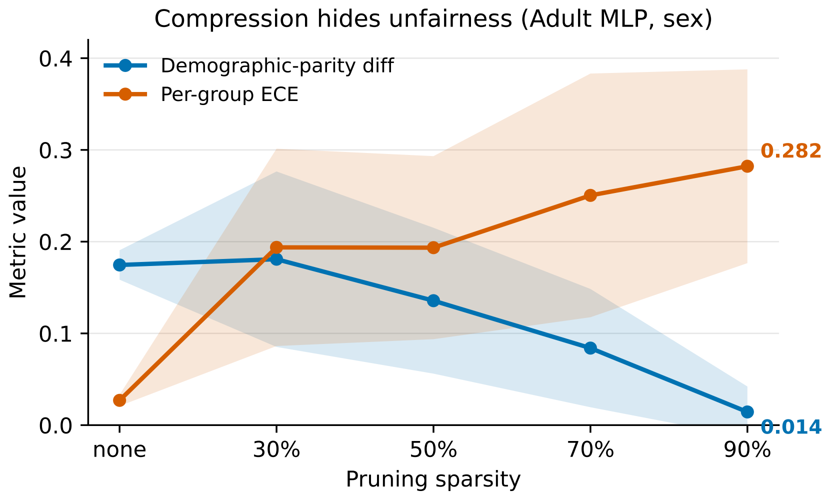

Figure 2: Finding 2. Adult MLP by sex under magnitude pruning. Demographic-parity difference falls while per-group ECE climbs; bands are ± 1 std over 10 seeds. The bands overlap at intermediate sparsities, so the robust claim is the endpoint contrast, not the middle ordering.

(0.081 ± 0.024). By calibration the ordering is different: White is only second-most-selected but by far the best calibrated (ECE 0.008 ± 0.002, three to seven times lower than every other group), while American-Indian-Eskimo is worst calibrated (0.056 ± 0.019). The smaller groups carry higher, noisier calibration error that the selection view never surfaces.

Adult by sex. The gap is large in selection rate (Male 0.253 ± 0.007 vs Female 0.079 ± 0.004) but negligible in calibration (ECE 0.009 ± 0.003 vs 0.014 ± 0.004). The point view and the calibration view disagree again, this time in the opposite way: a big point disparity with no calibration disparity.

COMPAS by race (positive outcome = flagged high-risk; higher is worse; groups with $n < 10$ dropped). African-American defendants are flagged at roughly twice the rate of the next group $(0.441 \pm 0.015$ vs Hispanic $0.249 \pm 0.046)$, the well-known point disparity. Yet the worst-calibrated group, Other (ECE 0.094 ± 0.029), is the one flagged *least* (0.164 ± 0.033), and the best-calibrated group is Caucasian (0.035 ± 0.016). The decoupling holds on a second dataset whose favorable direction is inverted, so it is not an Adult-specific label-convention artifact. The same pattern appears by sex: males are flagged more $(0.362 \pm 0.008$ vs $0.223 \pm 0.018)$ but females are markedly worse calibrated (ECE 0.085 ± 0.016 vs 0.033 ± 0.005).

Why it matters on-device. Sensitive labels are rarely available at inference time on a deployed device, so on-device audit triggers lean on confidence scores as a proxy [10]. If the two lenses disagree about which group is worst served, a confidence threshold applied uniformly across groups can track the wrong disparity, consistent with the observation that uniform abstention can amplify imbalance [17] and that point metrics can mask biased uncertainty [13]. Reporting both lenses is the cheap fix, and TinyAudit reports both by construction.

5 Result 2: compression can hide unfairness

We prune the Adult MLP to increasing sparsity and re-run the full audit at each level (Figure 2). By sex, the demographic-parity difference falls from 0.175 ± 0.016 (unpruned) to 0.014 ± 0.028 at 90% sparsity, a roughly 12-fold drop, while per-group ECE climbs from 0.027 ± 0.006 to 0.282 ± 0.106, a roughly 10-fold rise. By race the shape is the same: DP $0.150 \pm 0.024 \rightarrow 0.019 \pm 0.038$, ECE $0.051 \pm 0.007 \rightarrow 0.314 \pm 0.108$. The model looks fairer on parity precisely as its calibration collapses.

The mechanism is degradation toward a near-constant predictor, and we should be clear that the endpoint of that mechanism is textbook. A constant classifier trivially satisfies demographic parity, which is the standard counterexample for why parity alone is insufficient [16]. Our point is not that this

Table 1: Audit pipeline footprint (the pipeline itself, not the audited model), profiled with `tracemalloc` over 3 seeds. Peak RAM is governed by the streaming batch size, not the test-set size.

Dataset	Model	Batch	Audit peak RAM (KB)	Audit wall-clock (s)
Adult	logreg	64	610	0.03
Adult	logreg	256	641	0.06
Adult	logreg	1024	2,477	0.11
Adult	logreg	4096	9,826	1.10
Adult	MLP	256	946	0.08
Adult	MLP	4096	9,824	1.28
COMPAS	logreg	256	609	0.01
COMPAS	MLP	1543	1,181	0.05

endpoint exists but that a routine compression pipeline walks a model toward it gradually and silently, and that the audit metric most commonly reported improves monotonically the whole way. Nothing in a parity-only report distinguishes a model that became fairer from one that stopped discriminating because it stopped predicting. The corner case makes the endpoint vivid: the COMPAS logistic model at 90% pruning (only about 5 features to begin with) predicts a single class for everyone and reports DP $= 0.000$, DI $= 1.000$, perfect parity by construction. We read those cells as an illustration of the mechanism, not a trend.

Two qualifications. First, intermediate pruning levels are seed-noisy (parity std of 0.06 to 0.10), so the robust claim is the direction between endpoints. Second, the effect is strongest where pruning actually damages the model. The logistic model barely degrades (DP $0.174 \pm 0.008 \rightarrow 0.126 \pm 0.014$, ECE $0.012 \pm 0.002 \rightarrow 0.026 \pm 0.004$ at 90%), and correspondingly shows only a mild version of the effect. We also observe that int8 quantization can wreck calibration on its own; because int8 models' float weights are unreachable, their uncertainty columns are reported as missing rather than silently approximated, and temperature scaling on the audit set is a plausible mitigation.

The upshot for auditors: an audit that checks only demographic parity, run only after compression, would report the pruned model as the *fairest* in the sweep. An uncertainty-aware audit flags the same model as the most broken.

6 Feasibility: the audit fits on the device

Audit tooling is normally profiled not at all: the model's footprint is reported and the auditor's is assumed free because it runs on a workstation. If the audit is to run where the model runs, that assumption has to be checked. We profile the full three-method audit (fairness, uncertainty, explainability) with `tracemalloc` and report the pipeline's own peak RAM and wall-clock, not the audited model's (Table 1).

The working set is dominated by per-sample buffers: the feature matrix, the ensemble's stacked prediction arrays, and the explainer's perturbation batch. Fitting the Adult endpoints (625,005 bytes at batch 64 to 10,061,660 bytes at batch 4096) gives

$$\text{peak RAM} \approx 0.6\,\text{MB} + 2.3\,\text{KB} \times \text{batch size}. \tag{1}$$

The consequence is the structural claim, and it is the one we would defend: an arbitrarily large test set can be streamed through the audit in fixed-size batches at constant peak RAM. Audit cost is set by the batch size the engineer chooses, not by how much test data exists.

The absolute numbers need more care than the structure does. At a 256-row streaming batch the audit peaks at 0.61 to 0.95 MB and completes in under 0.1 s. That does *not* fit the 32 to 512 KB SRAM parts at the low end of the TinyML range, and we do not claim it does. It lands within reach of the high end, such as Cortex-M7 parts with roughly 1 MB of on-chip SRAM, and smaller batches move it further down: at batch 64 the Adult logistic audit peaks at 610 KB, and the fitted model in Equation 1 implies the per-sample term, not the fixed term, is what the engineer controls. Driving the roughly 0.6 MB fixed cost below 512 KB is the concrete engineering target this measurement identifies, and it is not yet met.

Two further honesty notes. The audited Adult MLP itself peaks at 6.1 MB, well above any microcontroller budget, so in this study it stands in for an MCU-class model rather than being one; the COMPAS logistic model at 62 KB is the realistic case. And `tracemalloc` excludes the CPython interpreter while counting Python object overhead the compiled on-device implementation would not pay, so the figure is neither a clean lower nor a clean upper bound on a C port. An actual MCU port is future work. What survives all of this is the scaling law, not the constant.

7 Limitations

Features are standardized before fitting, so the numbers describe the scaled-input regime. The uncertainty ensemble is five lightly-jittered copies of the one compressed model, not five from-scratch retrains, so uncertainty magnitudes are relative rather than absolute. int8 models' float weights are unreachable, so their uncertainty columns are blank, and decision trees have no dense weights and sit out the compression grid. Heavily-pruned COMPAS degenerates to a near-constant predictor; those cells illustrate a mechanism, not a trend. The footprint numbers exclude the interpreter, are measured on x86 rather than on a microcontroller, and at every batch size we measured they exceed the 512 KB SRAM of the smaller TinyML parts, so the streaming property rather than the absolute constant is what we claim. Neither of our two results is theoretically novel; both are predicted by prior work [11, 4, 9], and the contribution is measuring them on the compressed model inside the audit's own budget. Exact reproducibility is guaranteed on Linux/x86; elsewhere expect last-digit float drift, which the reported std columns absorb.

8 Conclusion

Fairness toolkits assume the cloud [14, 2, 8]; TinyML benchmarks ignore fairness [19]; uncertainty now fits on tiny devices [7] but has not been pointed at fairness there. TinyAudit couples the three axes and measures them on the compressed artifact rather than the full-precision one. The theory predicting our two results predates us [11, 4, 9]; what we add is that the predicted effects are measurable, group-selective, and actively misleading on the model that actually ships, and that measuring them costs under a megabyte at a streaming batch, constant in test-set size. The practical warning is narrow and, we think, hard to dismiss: an audit that reports demographic parity alone, run after compression, will rank the most degraded model in a pruning sweep as the fairest one. Closing the remaining gap to a 512 KB part, and porting the pipeline to real hardware, is the next step rather than a solved problem. All numbers regenerate from committed CSVs, and every figure is produced from the same files the tables cite.

References

[1] J. Angwin, J. Larson, S. Mattu, and L. Kirchner. Machine bias. *ProPublica*, 2016.

[2] W. H. Deng et al. Exploring how machine learning practitioners (try to) use fairness toolkits. In *Proc. ACM Conference on Fairness, Accountability, and Transparency (FAccT)*, 2022.

[3] C. Blakeney, N. Huish, Y. Yan, and Z. Zong. Simon says: Evaluating and mitigating bias in pruned neural networks with knowledge distillation. arXiv:2106.07849, 2021.

[4] A. Chouldechova. Fair prediction with disparate impact: A study of bias in recidivism prediction instruments. *Big Data*, 5(2):153–163, 2017.

[5] F. Ding, M. Hardt, J. Miller, and L. Schmidt. Retiring Adult: New datasets for fair machine learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.

[6] A. Fabris, S. Messina, G. Silvello, and G. A. Susto. Algorithmic fairness datasets: the story so far. *Data Mining and Knowledge Discovery*, 36(6):2074–2152, 2022.

[7] N. Ghanathe and S. Wilton. QUTE: Quantifying uncertainty in TinyML models with early-exit-assisted ensembles for model-monitoring. In *Proc. International Conference on Machine Learning (ICML)*, PMLR vol. 267, 2025.

[8] X. Han, J. Chi, Y. Chen, Q. Wang, H. Zhao, N. Zou, and X. Hu. FFB: A fair fairness benchmark for in-processing group fairness methods. In *Proc. International Conference on Learning Representations (ICLR)*, 2024.

[9] S. Hooker, N. Moorosi, G. Clark, S. Bengio, and E. Denton. Characterising bias in compressed models. arXiv:2010.03058, 2020.

[10] P. J. Kenfack, S. E. Kahou, and U. Aïvodji. Fairness under demographic scarce regime. *Transactions on Machine Learning Research (TMLR)*, 2024.

[11] J. Kleinberg, S. Mullainathan, and M. Raghavan. Inherent trade-offs in the fair determination of risk scores. In *Proc. Innovations in Theoretical Computer Science (ITCS)*, 2017.

[12] R. Kohavi and B. Becker. UCI Adult (Census Income) data set. UCI Machine Learning Repository, 1996.

[13] S. Kuzucu, J. Cheong, H. Gunes, and S. Kalkan. Uncertainty as a fairness measure. *Journal of Artificial Intelligence Research (JAIR)*, 2024.

[14] M. S. A. Lee and J. Singh. The landscape and gaps in open source fairness toolkits. In *Proc. CHI Conference on Human Factors in Computing Systems*, 2021.

[15] T. Nguyen, D. Nguyen, and H. Cao. XEdgeAI: A human-centered industrial inspection framework with data-centric explainable edge AI approach. *Information Fusion*, vol. 116, 2024.

[16] G. Pleiss, M. Raghavan, F. Wu, J. Kleinberg, and K. Q. Weinberger. On fairness and calibration. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

[17] L. Rosenblatt and R. T. Witter. FairlyUncertain: A comprehensive benchmark of uncertainty in algorithmic fairness. arXiv:2410.02005, 2024.

[18] A. Salih, I. B. Galazzo, P. Gkontra, et al. A perspective on explainable artificial intelligence methods: SHAP and LIME. *Advanced Intelligent Systems*, 2025.

[19] S. Somvanshi, M. Islam, S. Chhetri, S. Chakraborty, et al. From tiny machine learning to tiny deep learning: A survey. *ACM Computing Surveys*, 2025.